NB-AE-222: Gas phase H2 measurements with Unisense H2 sensor, GC calibration

Date: 2025-05-16
Tags: Test AE Calibration Future NB GC H2 Gas chromatography Unisense H2 Sensor In situ
Category: SrTiO3
Status: Done
Created by: Nadzeya Brezhneva

Literature/reference experiments

Literature	/
Reproduction	/
Similar experiments	SrTiO3 - AE-483: Testing of in-situ electrochemical hydrogen sensor and test H2 evolution reaction with RhCrO:Al:SrTiO3 SrTiO3 - AE-484: Testing of in-situ electrochemical hydrogen sensor and test H2 evolution reaction with RhCrO:Al:SrTiO3 SrTiO3 - NB-AE-215 / AE-485: Gas phase H2 measurements with H2 sensor, SrTiO3:Al loaded with RhxCryO3 (NB-162-4) suspension (1 mg/ml), 365 nm UHP LED, check with GC measurements

Reagents

Name	CAS Number / Experiment Number	Inventory number	Amount [mmol]	Equivalents	$Mass_{theo}$ [mg]	$Mass_{exp}$ [mg]	Molar mass [g/mol]	Density (g/ml)	Volume [ml]	pressure [bar]
milli-Q H_2O	/	/	/	/	/	/	/	/	ca. 5 (for H2 saturated atmosphere in the RBF)	/
Hydrogen	1333-74-0	/	/	/	/	/	/	/	1 ballon (approx. 2 L)	approx. 1

Procedure/observations

Date	Time	Step	Observations	Pictures
21.05.2025	ca. 10:10	The Device was connected to the computer and the device was conected to the sensor and the PT100.The experiment was started on the device - the PreAmp was set at 1pA=1mV and the polarisation was 100 mV	/	/
	ca. 10:25	The logging was started (sample interval for pre-measurements was set 10 s).	**NB-AE-222-Logger1**	/
	10:25	Attempt to connect PT 100 sensor.	Problems with recognizing PT 100, by now no use of the sensor.	/
	ca. 10:30	A 10 mL RBF flask was filled with Milli-q water (approx. 5 mL) (flask 1)	/	20250521_104013-RBF-with-H2O.jpg
	ca. 10:45-50	Filling the balloon with H_2 gas in E005 lab according to the protocol Protocol - Getting hydrogen from hydrogen bottle in CEEC II E014	/	/
	ca.10:50	Start purging flask 1 with H_2 from the balloon (two cannulas in the septum) according to the protocol Protocol - Gas phase calibration of H2 UniAmp sensor.	/	20250521_105118-H2-purging-with-balloon.jpg
	ca. 11:05	Important note: removing the second needle to reduce too fast release of H_2 from the balloon.	/	/
	11:10-12:40	Continuous recording during pre-measurements were continued.	/	Unisense-pre-measurements-Screenshot-2025-05-21-160016.png

	12:40	PT 100 sensor was connected.	Now everything went well, the sensor is recognized.	/
	12:46	The logging was stopped	/	/
	ca. 12:50	The H2 sensor was placed on a Gl14/NS14 5 mL Schlenk flask (flask 2) equipped with a PTFE stirring bar insisde (Prep work - AE-482: Volume determination of GL14/NS14 5 mL Schlnek flask) using a 2.0 mm BOLA fitting	/	20250521_130448-GL14-NS14-flask-setup.jpg 20250521_130207-with-H2-sensor.jpg
	13:02	Start stirring (400 rpm), LAUDA on (25 °C), PT 100 sensor was placed inside the water bath.		
	13:06	A new logging was started.	**NB-AE-222-Logger2**	/
	13:12	The first calibration point was taken at 0 ppm H2	Calibration --> Offset --> Set value close to 0 (in current experiment -4 mV).	Unisense-calibration-Screenshot-2025-05-21-160445.png
		Calculations of the amount of saturated H_2 gas that is needed to be injected into a GL14/NS14 flask *V_{total} (GL14/NS14) = 11.66 mL according to Prep work - AE-482: Volume determination of GL14/NS14 5 mL Schlnek flask* *0.05 % (V_{total}) = 0.0058 mL = 5.8 µL **(6 µL)*** *0.3 % (V_{total}) = 0.0348 mL = 34.8 µL (**35 µL**)* *1% (V_{total}) = 0.116 mL = **116 µL***		
	13:25	6 uL of H_2 saturated gas was taken from the head space of flask 1 using 100 µL Pressure lok syringe and injected into flask 2 (via septum on the NS14 neck).	/	/
	13:25-27	Waiting for ca. 1-2 min for stabilizing the signal, afterwards adding point to the calibration curve (corresponded to the signal 2.8 mV).		
	13:29	Taking 100 uL of the sample from the head space of GL14/NS14 flask (preliminary flushing it with the gas phase from the head space) and kept in Pressure lok syringe for further investigations with GC.	**NB-AE-222-gas 0.05 %theor H2**	Unisense-signal-vs-time-Screenshot-2025-05-21-155932.png
	13:30-35	The sample was transferred to GC lab (314) in TO Building.		
	ca. 13:50	GC measurements of the sample **NB-AE-222-gas 0.05 %theor H2** with Equipment - GC chromatograph Schimadzu Nexis GC-2030, Peneva`s Lab 304, Physical Chemistry building in accordance with the protocol Protocol - Hydrogen Measurement Using GC.	Change of septum in the GC --> delay in start of the measurements in ca. 15 min	/
	14:07	For 0.3 vol.% H_2 in total volume, 29 uL H_2 saturated gas was taken from the head space of flask 1 using Pressure lok syringe and injected into flask 2 (via septum on the NS14 neck).	Since the signal after first injection of H2 sat gas was stable, it was necessary to inject just 29 uL (35 uL-6 uL) instead of the required 35 uL	/
		Important note: adjust the parameters in H_2 table each time before injecting the sample (check current values of temperatute 24.5°C and pressure 991 mbar). The input of the value in ppm will give the result in Pa at the bottom of the panel.	Screenshot-2025-05-27-184655-Unisense calculator.png	/
	14:08	Waiting for 1 min for stabilizing signal, addition of the third point to the calibration curve.	/	/
	ca. 14:10	Taking 100 uL of the sample from the head space of GL14/NS14 flask (preliminary flushing it with the gas phase from the head space) and kept in Pressure lok syringe for further investigations with GC.	**NB-AE-222-gas 0.3 %theor H2**	Unisense-signal-vs-time-Screenshot-2025-05-21-155932.png
	ca. 14:10-15	The sample was transferred to GC lab (314) in TO Building.	/	/
	ca. 14:20	GC measurements of the sample **NB-AE-222-gas 0.3 %theor H2** with Equipment - GC chromatograph Schimadzu Nexis GC-2030, Peneva`s Lab 304, Physical Chemistry building in accordance with the protocol Protocol - Hydrogen Measurement Using GC.	/	/
	ca. 14:30	For 1 vol.% H_2 in total volume, 82 uL H_2 saturated gas was taken from the head space of flask 1 using Pressure lok syringe and injected into flask 2 (via septum on the NS14 neck).	Since the signal after the second injection of H2 sat gas was stable, it was necessary to inject just 82 uL (117 uL-35 uL) instead of the required 117 uL	/
	ca. 14:32	Waiting for ca. 1-2 min for stabilizing signal, addition of the fourth point to the calibration curve.	/	/

	ca. 14:35	Taking 100 uL of the sample from the head space of GL14/NS14 flask (preliminary flushing it with the gas phase from the head space) and kept in Pressure lok syringe for further investigations with GC.	**NB-AE-222-gas 1 %theor H2**	Unisense-signal-vs-time-Screenshot-2025-05-21-155932.png
	ca. 14:40	The sample was transferred to GC lab (314) in TO Building.	/	/
	ca. 14:50	GC measurements of the sample **NB-AE-222-gas 1 %theor H2** with Equipment - GC chromatograph Schimadzu Nexis GC-2030, Peneva`s Lab 304, Physical Chemistry building in accordance with the protocol Protocol - Hydrogen Measurement Using GC.	/	/
	ca. 15:20	The calibration was applied	/	/
	15:22	The logging was stopped and subsequently the experiment was stopped	/	
	ca. 15:40	Deassembling the setup.	/	/
22.05.2025	ca. 15:40	GC calibration using 25 uL, 50 uL, 75 uL, 100 uL H_2 with 1 % All-in-gas reference gas mixture (1 vol.%) and Pressure lok syringe using Equipment - GC chromatograph Schimadzu Nexis GC-2030, Peneva`s Lab 304, Physical Chemistry building in accordance with the protocol Protocol - Hydrogen Measurement Using GC.	/	/

Analysis

Date	Time	Sample name	Analysis method	Analytical device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
21.05.2025	10:25-12:46	NB-AE-222-Logger1	electrochemical H2 detection	Equipment - H2 UniAmp Sensor	/	NB-222-1.ulog NB-AE-222-log1-pre-measurements.xlsx Calibration-data-for-pre-measurements.csv Comments-pre-measurements.csv Data-1-H2-UniAmp-503157-pre-measurements.csv Devices-pre-measurements.csv Miscellaneous-pre-measurements.csv	Unisense-pre-measurements-Screenshot-2025-05-21-160016.png	/	Pre-measurements are done, the signal is stable during measurements.
	13:06-15:35	NB-AE-222-Logger2	electrochemical H2 detection	Equipment - H2 UniAmp Sensor	/	NB-222-1.ulog NB-AE-222-log2-calibration.xlsx Calibration-data-for-calibration.csv Comments-for-calibration.csv Data-1-H2-UniAmp-503157-for-calibration.csv Devices-for-calibration.csv Miscellaneous-for-calibration.csv	Unisense-calibration-Screenshot-2025-05-21-160445.png Unisense-signal-vs-time-Screenshot-2025-05-21-155932.png Unisense-calculations.xlsx	/	*Signal vs. time* Step-by-step increase in signal value after gradual injection of H_2-containing mixture. *4-point calibration* R^2 = 1, intercept = -1.031, slope = 0.078. After processing Unisense data: theor 0.05 vol.% ---> 0.0517 vol.% theor 0.3 vol.% ---> 0.3001 vol.% theor 1 vol.% ---> 1.0004 vol.%
	13:29	NB-AE-222-gas 0.05 % theor H2	GC	Equipment - GC chromatograph Schimadzu Nexis GC-2030, Peneva`s Lab 304, Physical Chemistry building	/	Calibration-Unisense.gcb Blank1.gcd theor-0.05-vol.gcd theor-0.05-vol-H2-GC.txt	theor-0.05-vol.-H2.PNG	/	Rough estimation by LabSolution software (conc vol.%) 0.239, much higher than expected
	14:10	NB-AE-222-gas 0.3 % theor H2	GC	Equipment - GC chromatograph Schimadzu Nexis GC-2030, Peneva`s Lab 304, Physical Chemistry building	/	Calibration-Unisense-1.gcb Blank1.1.gcd theor-0.3-vol.gcd theor-0.3-vol-H2-GC.txt	theor-0.3-vol.-H2.PNG	/	Rough estimation by LabSolution software (conc vol.%) 0.569, higher than expected
	14:35	NB-AE-222-gas 1 % theor H2	GC	Equipment - GC chromatograph Schimadzu Nexis GC-2030, Peneva`s Lab 304, Physical Chemistry building	/	Calibration-Unisense-2.gcb Blank1.2.gcd theor-1-vol.gcd theor-1-vol-H2-GC.txt	theor-1-vol.-H2.PNG	/	Rough estimation by LabSolution software (conc vol.%) 1.122, a bit higher, but comparable with expected value
22.05.2025	15:40	/	GC	Equipment - GC chromatograph Schimadzu Nexis GC-2030, Peneva`s Lab 304, Physical Chemistry building	/	NB-222-calibration.gcb Blank002.gcd 100H2-calibration-GC.gcd 100H2-calibration-GC.txt 75H2-calibration-GC.gcd 75H2-calibration-GC.txt 50H2-calibration-GC.gcd 50H2-calibration-GC.txt 25H2-calibration-GC.gcd 25H2-calibration-GC.txt	100-H2.PNG 75-H2.PNG 50-H2.PNG 25-H2.PNG GC_Data_Analysis_H2-1.py NB_222_Data.csv NB_222_Data-Results.txt NB_222_Data-Analysis.tiff	/	After calibration and processing data in Python: according to GC: theor 0.05 vol.% ---> 0.252 vol.% theor 0.3 vol.% ---> 0.588 vol.% theor 1 vol.% ---> 1.151 vol.%

Results

H_2 test measurements with Unisense hydrogen sensor was performed. GC measurements (as well as 4-point calibration with 1 vol.% H_2 gas mixture) was performed as a control.

Four points were tested - 0, 0.05 vol.%, 0.3 vol.%, 1 vol.% (theoretical values)

According to the Unisense data:

theor 0.05 vol.% ---> 0.0517 vol.%

theor 0.3 vol.% ---> 0.3001 vol.%

theor 1 vol.% ---> 1.0004 vol.%

Used as calibration data for Unisense.

Check with GC (after 4 point calibration with 1 vol.% H_2 gas mixture and processing data in Python):

theor 0.05 vol.% ---> 0.252 vol.%

theor 0.3 vol.% ---> 0.588 vol.%

theor 1 vol.% ---> 1.151 vol.%

Unisense data are in a very good agreement with theoretical values, whereas GC data (especially theor 0.05 vol.% and 0.3 vol.%) are not in good agreement with Unisense and theoretical values, high error.

Future recommendations

Old procedure	Problem	Future recommendations
/	/	placing water into Schlenk where H2 measurement is performed.
/	/	In future: exporting only data-1 csv file from Unisense Logger software.

Linked experiments

Prep work - AE-477: Testing of in-situ electrochemical hydrogen sensor and test H2 evolution reaction with RhCrO:Al:SrTiO3

Prep work - AE-482: Volume determination of GL14/NS14 flask

SrTiO3 - NB-162: Preparation of SrTiO3:Al with RhyCr2-yO3 (Osterloh route), I attempt, 1000 C 10h

SrTiO3 - AE-483: Testing of in-situ electrochemical hydrogen sensor and test H2 evolution reaction with RhCrO:Al:SrTiO3

SrTiO3 - AE-484: Testing of in-situ electrochemical hydrogen sensor and test H2 evolution reaction with RhCrO:Al:SrTiO3

SrTiO3 - NB-AE-215 / AE-485: Gas phase H2 measurements with H2 sensor, SrTiO3:Al loaded with RhxCryO3 (NB-162-4) suspension (1 mg/ml), 365 nm UHP LED, check with GC measurements

Linked resources

Equipment - GC chromatograph Schimadzu Nexis GC-2030, Peneva`s Lab 304, Physical Chemistry building

Equipment - H2 UniAmp Sensor - Low range - 2.1 x 80 mm needle

General Protocol - Hydrogen Measurement Using GC (AG Paneva)

General Protocol - Getting hydrogen from hydrogen bottle in CEEC II E014

General Protocol - Gas phase calibration of H2 UniAmp sensor

Light Source - UHP LED 365 nm-1

Attached files

Screenshot-2025-05-27-184655-Unisense-calculator.png
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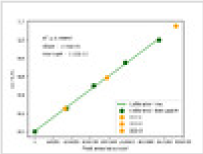

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GC_Data_Analysis_H2-1.py
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Unisense-calculations.xlsx
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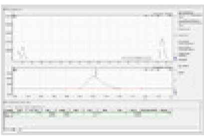

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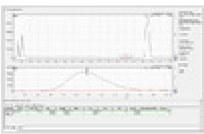

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NB-222-calibration.gcb

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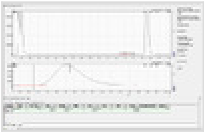

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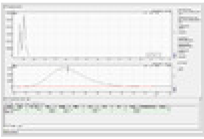

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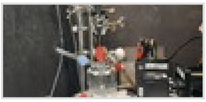

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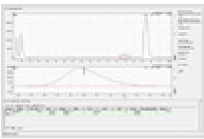

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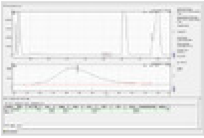

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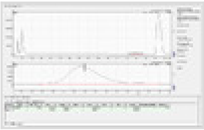

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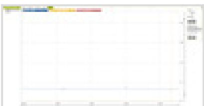

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Comments-pre-measurements.csv
sha256: 4aa81964d195af15a2cb230da0cdf929bf3fa1e8b671e563972a03a18fb98067

Comments-for-calibration.csv
sha256: 4aa81964d195af15a2cb230da0cdf929bf3fa1e8b671e563972a03a18fb98067

Calibration-data-for-calibration.csv
sha256: 568030a0e4c29c04e6cb8120d3f500f2c3ca3e7e492a8d6ae1e3363511a608cb

Calibration-data-for-pre-measurements.csv
sha256: 568030a0e4c29c04e6cb8120d3f500f2c3ca3e7e492a8d6ae1e3363511a608cb

NB-AE-222-log2-calibration.xlsx
sha256: 2181dea31ff813012ec1dbc581294469f17c9ab664bbf76cf4474613bb13e03b

NB-AE-222-log1-pre-measurements.xlsx
sha256: 11649e79b8f56d1e6d12c188eb548fae762688874a7035e9a4d7ef6a2c8ec05e

NB-222-1.ulog
sha256: 139fb27706a12def8b7eb712d787bc97040858e8d1a715257bfe504c3a136cbc

Comments

On 2025-05-27 12:14:38 Jacob Schneidewind wrote:
Notes:
* Adding "in situ" tag
* Removing literature
* Linking H2 sensor calibration protocol (gas phase)
* Linking protocol for how to fill H2 balloon
* Specifying where PT100 was placed, ideally also including a picture with Pt100 in the future
* 13:25: specifying that 100 ul pressure lok syringe is used
* Linking protocol for measurements of GC samples
* 14:07: including screenshot of H2 table in software, explaining this in a bit more detail
* In the future only exporting data-1 csv file
* Analysis table: splitting data for log 1 and log 2

* Future recommendation: placing water into Schlenk where H2 measurement is performed (also adjusting this in general protocol)

On 2025-05-27 19:05:59 Nadzeya Brezhneva wrote:
Notes:
* Adding "in situ" tag -> done
* Removing literature -> done
* Linking H2 sensor calibration protocol (gas phase) -> done
* Linking protocol for how to fill H2 balloon -> done
* Specifying where PT100 was placed, ideally also including a picture with Pt100 in the future -> done, ok, the note will be taken into account
* 13:25: specifying that 100 ul pressure lok syringe is used -> done
* Linking protocol for measurements of GC samples -> done
* 14:07: including screenshot of H2 table in software, explaining this in a bit more detail -> done
* In the future only exporting data-1 csv file -> done, it will be taken into account
* Analysis table: splitting data for log 1 and log 2 -> done

* Future recommendation: placing water into Schlenk where H2 measurement is performed (also adjusting this in general protocol) --> done

Unique eLabID: 20250516-d194458a77d779c83a79b1695c2138f354aa148e
Link: https://elab.water-splitting.org/experiments.php?mode=view&id=2173